

Bayesian Networks, Influence Diagrams & Association Rules

Graph-Based Probabilistic Reasoning and Unsupervised Pattern Discovery

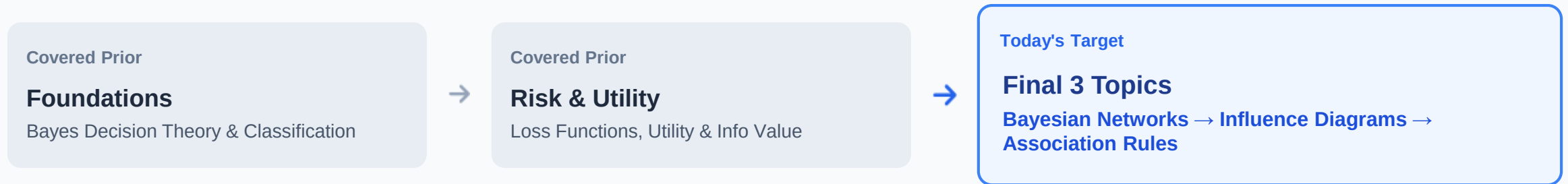
Prescribed Text: Ethem Alpaydin (Ch. 3)

University Exam Focus

TEACHER & STUDENT NOTE

Welcome students to the concluding lecture of Bayesian Decision Theory. Explain that today moves from single-formula decision rules to graphical networks and transactional pattern mining.

Unit II Progression: The Final Step



TEACHER & EXAM NOTE

Tell students: "We have already mastered calculating risks and posteriors. Today we learn how multiple random variables connect graphically without massive joint probability tables."

What is a Bayesian Network?

"A Bayesian Network is a graphical representation of probabilistic relationships between variables."

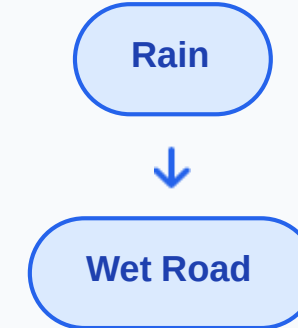
Graph Structure: Directed Acyclic Graph (DAG) with no loops.

Nodes: Represent random variables (discrete or continuous).

Directed Edges: Represent probabilistic dependencies or cause-effect.

Probabilities: Describe the strength of belief under uncertainty.

VISUAL INTUITION



Rain causes road wetness. Edge indicates conditional dependency.

SPEAKER & EXAM CUE

Exam 2-Marker: Students must state both components: *Directed Acyclic Graph (DAG)* + *Conditional Probability Tables (CPT)*.

Why Do We Need Bayesian Networks?

THE BOTTLENECK

Full Joint Distribution Fails

For n binary variables, storing all probabilities requires $2^n - 1$ parameters. For 30 variables, that is over 1 billion values!

Result: Equations become intractable to calculate and store.

THE SOLUTION

Sparse Graph Decomposition

Real-world variables depend only on their direct causes. Bayesian Networks exploit local independence so reasoning requires drastically fewer values.

Variables > Relationships > Local CPTs > Visual Inference

TEACHER'S ANALOGY

Tell students: "Instead of writing a 1,000-page book of every possible combination in life, a Bayesian network only writes down direct cause-effect rules."

Components of a Bayesian Network



1. Nodes (Variables)

Each node represents an observable or unobservable random variable.

Example: *Disease, Fever, Test Result*



2. Directed Edges

Arrows show direct dependencies or causal influences between nodes.

Example: *Disease → Symptoms*



3. Conditional Tables

Each node has a CPT specifying $P(\text{Node} \mid \text{Parents})$ to quantify influence.

Example: $P(\text{Fever} \mid \text{Disease} = \text{True})$

EXAM NOTE

Important for 5 Marks: Diagram must be an **Acyclic** graph. Mention: "No directed cycle or feedback loop is permitted in a Bayesian Network."

Beginner-Friendly Medical Example

Disease (Flu)

Fever

Cough

Diagnostic Reasoning (Inference)

In practice, doctors observe the symptoms first (Fever, Cough) and want to infer the hidden cause (Disease).

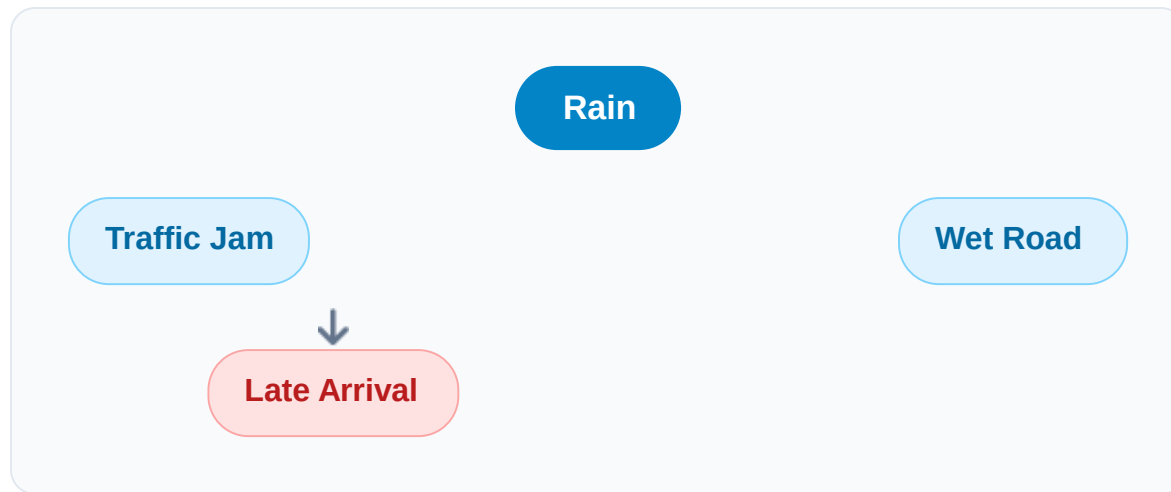
$P(\text{Disease} \mid \text{Fever, Cough})$

Calculated by Bayes' Rule using the conditional tables of Fever and Cough!

CLASSROOM ACTIVITY

Ask students: "If a patient walks in with cough and fever, are these symptoms independent? No! But once we know they have Flu, fever and cough become conditionally independent."

Multi-Node Network Architecture



Arrow Semantics (What arrows mean)

Rain → Traffic: Rain increases the probability of traffic delays.

Traffic → Late Arrival: Traffic directly increases probability of arriving late.

Rain → Wet Road: Wet road does not cause late arrival directly in this model.

EXAM TIP

Draw this exact 4-node diagram for 5-mark university questions on "Explain Bayesian Networks with a suitable example."

Understanding Conditional Independence

"Two variables X and Y are conditionally independent given Z if knowing Z makes X tell us nothing new about Y."

Mathematical Statement:

$$P(X, Y | Z) = P(X | Z) \cdot P(Y | Z)$$

Plain English Translation: Once Z is known, the joint probability simplifies into the product of individual probabilities.

Example: Fever & Cough

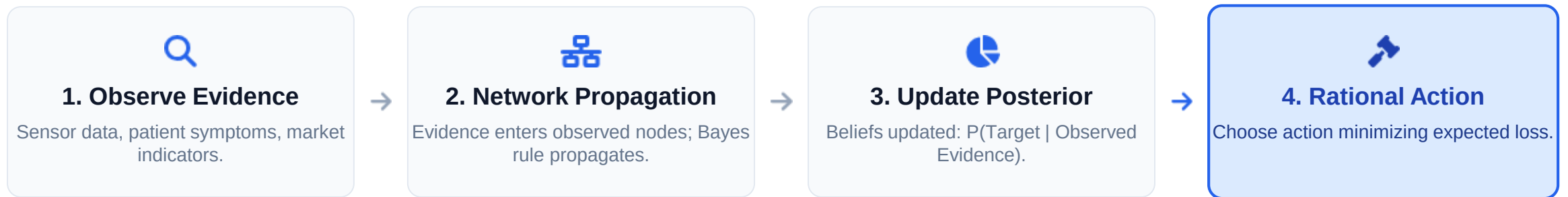
Normally, Fever and Cough occur together. But once we confirm **Flu**, knowing about Fever gives no extra information about Cough.

Fever $\perp$ Cough | Flu

EXAM NOTE

Define: "Conditional independence is the primary property that allows Bayesian Networks to compress massive probability tables into simple local models."

How Bayesian Networks Aid Decisions



CONNECTING PAST CLASSES

Remind students: In Class 1 we learned Bayes' rule for 1 feature. Bayesian Networks generalize Bayes' rule to hundreds of interacting features!

Bayesian Network: Exam Revision Card

Official Exam Definition

"A Bayesian Network is a Directed Acyclic Graph (DAG) whose nodes represent random variables and directed edges represent conditional dependencies, parameterized by Conditional Probability Tables."

Memory Keyword: DAG + CPT = Bayesian Network

KEY PROPERTIES TO STATE

- ✓ Directed Acyclic Graph (DAG)
- ✓ Encodes Conditional Independence
- ✓ Handles Incomplete / Missing Evidence
- ✓ Enables Bidirectional Inference (Cause ↔ Effect)

SPEAKER CUE

Conclude Section 1. Ask students if they understand why edges are one-way (directed) and why cycles are forbidden before proceeding to Influence Diagrams.

What is an Influence Diagram?

"An Influence Diagram is a graphical model that represents uncertainty, decisions, and utilities together in a single framework."

It directly extends a Bayesian Network by adding human agency (**Decisions**) and outcome values (**Utilities**).

Bayesian Network
Uncertainty Only

Influence Diagram
Uncertainty + Action + Payoff

THE CORE QUESTION

Bayesian Net asks:
"What is likely?"

Influence Diagram asks:
"What should we do?"

SPEAKER & EXAM CUE

2-Mark Question: "Define Influence Diagram." Emphasize the 3 key pillars: Chance, Decision, and Utility.

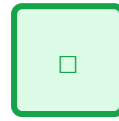
The Three Standard Node Types



Chance Node

SHAPE: CIRCLE / OVAL

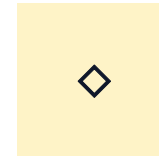
Represents uncertain random variables outside our direct control.



Decision Node

SHAPE: RECTANGLE

Represents choices or actions under the decision maker's control.



Utility Node

SHAPE: DIAMOND

Represents payoffs, costs, rewards, or preference satisfaction.

EXAM NOTE

Standard 5-Mark Question: "Draw and describe the three node shapes used in an Influence Diagram." Circles for Chance, Rectangles for Decisions, Diamonds for Utility.

1. The Chance Node (Circle ○)

Nature's Uncertain States

A chance node represents an uncertain event that the decision maker cannot directly force or choose.

Governed by Probability Distributions

Each chance node stores a probability table conditioned on parent nodes.

REAL-WORLD EXAMPLES

○ Patient Disease (Flu / Healthy)

○ Weather (Sunny / Rainy)

○ Market Demand (High / Low)

SPEAKER TIP

Remind students: "Every node that existed in a Bayesian Network was a Chance Node! That is why an Influence Diagram directly contains Bayesian Networks."

2. The Decision Node (Rectangle □)

The Agent's Controlled Action

A decision node represents an explicit choice from a set of mutually exclusive alternatives available to the agent.

No Probabilities Inside

Incoming arrows represent information known at decision time, NOT probabilistic causes.

REAL-WORLD EXAMPLES

□ **Medical Treatment (Drug A vs Drug B)**

□ **Loan Approval (Approve / Reject)**

□ **Marketing Action (Email / SMS / None)**

SPEAKER CUE

Highlight: "A doctor cannot choose whether a patient has a disease (that's Chance), but the doctor can choose which medication to prescribe (that's Decision)!"

3. The Utility Node (Diamond ◇)

Quantifying Desirability & Value

A utility node represents the payoff, cost, or benefit of every possible combination of decisions and outcomes.

Utility Axiom

Higher utility value = more desirable outcome. Rational decision makers maximize Expected Utility.

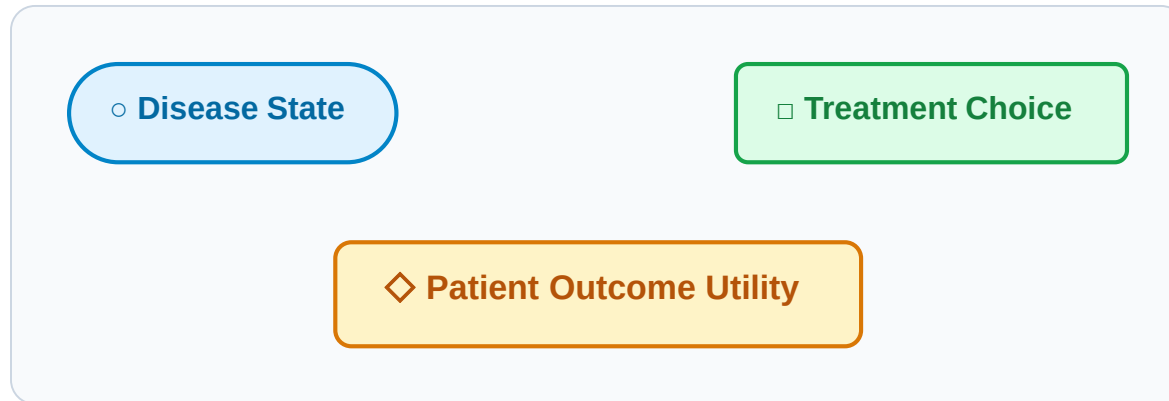
WHAT UTILITY REPRESENTS

- **Patient Health Score:** Cured = +100, Side Effect = -20
- **Financial Profit:** Revenue minus operational cost
- **Travel Time Saved:** On-time arrival vs Late penalty

PAST CLASS CONNECTION

Remind students: In Class 2 we studied Utility Functions $U(s)$. The Utility node simply evaluates $U(\text{Action}, \text{Outcome})$ to select the optimal choice.

Medical Influence Diagram Architecture



How the System Decides:

- 1. Assess Chance:** Probability of disease is computed from tests.
- 2. Simulate Actions:** Treatment options (Drug A vs Drug B) are evaluated.
- 3. Calculate Utility:** Weighted payoff combines cure rate and adverse costs.

CLASSROOM ACTIVITY

Ask students: "Why does Utility have arrows entering from BOTH Disease and Treatment?" Answer: Because utility depends on whether the patient was actually sick AND which treatment was administered!

Bayesian Network vs. Influence Diagram

Feature	Bayesian Network	Influence Diagram
Primary Goal	Probabilistic reasoning under uncertainty	Optimal decision-making under uncertainty
Node Types	Chance Nodes only (Circles)	Chance (○), Decision (□), Utility (◇)
Agent Control	No decision nodes (Passive observer)	Explicit decision nodes (Active agent)
Core Question	"What is the probability of X?"	"What action gives maximum utility?"

MAJOR EXAM 5-MARK QUESTION

Students should memorize the memory line: "Bayesian Network = What is likely? | Influence Diagram = What should we do?"

Influence Diagrams & Expected Utility

The Principle of Maximum Expected Utility

An influence diagram solves a decision problem by choosing the action a^* that yields highest expected utility across all uncertain outcomes.

$$EU(\text{Action}) = \sum P(\text{Outcome} \mid \text{Action}) \cdot \text{Utility}(\text{Outcome})$$

Probability comes from Chance nodes; Utility comes from Utility nodes.

DECISION PIPELINE

1. Enter Evidence

2. Update Chance Nodes

3. Calculate Expected Utility

4. Select Action with Max EU

SPEAKER TIP

Remind students that they already mastered this formula in Class 2 under Utility Theory. The Influence Diagram is simply the graphical tool executing it.

What Are Association Rules?

"Association rules are used to discover interesting relationships, co-occurrences, or patterns between items in large datasets."

They take the symbolic if-then format:

$X \rightarrow Y$ (Antecedent $\rightarrow$ Consequent)

 CRITICAL CAVEAT

Association $\neq$ Causation!

Buying Bread does not physically cause someone to buy Milk; they simply co-occur frequently in customer baskets.

EXAM NOTE

2-Marker: "Define Association Rule." Key words: *Unsupervised pattern discovery*, *co-occurring itemsets*, format $X \rightarrow Y$.

Market Basket Analysis: Sample Dataset

Transaction ID	Items Purchased
T1	Bread, Milk
T2	Bread, Butter
T3	Bread, Milk, Butter
T4	Milk, Butter

DISCOVERED RULE

Bread → Milk

"Customers who purchase Bread are very likely to also purchase Milk."

Total Transactions (N) = 4

Transactions with Bread = 3 (T1, T2, T3)

Transactions with Both = 2 (T1, T3)

CLASSROOM ACTIVITY

Point to this table on the board. Tell students: "We will now calculate Support, Confidence, and Lift directly on these 4 transactions."

Industry Applications of Association Rules



Supermarket Layout

Placing bread near milk to encourage bundled purchases or cross-sales.



E-Commerce Engines

"Frequently Bought Together" recommendations on Amazon and Flipkart.



Medical Diagnostics

Discovering adverse drug interactions and symptom co-occurrences.



Web Clickstreams

Analyzing webpage sequence visits to optimize user navigation paths.

EXAM NOTE

Students should write at least 2 real-world domains when answering the 5-mark question "Explain Association Rule Mining."

Support: How Common Is the Combination?

Support measures the proportion of total transactions in the dataset containing both itemsets X and Y.

$$\text{Support}(X \rightarrow Y) = \text{Count}(X \cup Y) / \text{Total Transactions (N)}$$

In probability notation: $P(X \cap Y)$

Memory Line: "Support = How common is the combination overall?"

NUMERICAL EXAMPLE

Total transactions = **100**

Transactions with (Bread + Milk) = **20**

$$\text{Support} = 20 / 100 = 20\% (0.20)$$

EXAM NOTE

2-Mark Question: "Define Support." Students must provide the formula and explain that rules with low support are usually discarded as rare noise.

Confidence: How Certain Is the Rule?

Confidence measures how often item Y is purchased among transactions that already contain item X.

$$\text{Confidence}(X \rightarrow Y) = \text{Count}(X \cup Y) / \text{Count}(X)$$

In probability notation: Conditional Probability $P(Y | X)$

Memory Line: "Confidence = When X happens, how often does Y happen?"

NUMERICAL EXAMPLE

Transactions containing Bread = 40

Transactions with (Bread + Milk) = 20

$$\text{Confidence} = 20 / 40 = 50\% (0.50)$$

EXAM NOTE

Warn students: Confidence is directional! Confidence(Bread $\rightarrow$ Milk) is generally NOT equal to Confidence(Milk $\rightarrow$ Bread).

Lift: How Strong Is the True Association?

Lift measures how much more often X and Y occur together than if they were totally statistically independent.

$$\text{Lift}(X \rightarrow Y) = P(X \cap Y) / [P(X) \cdot P(Y)]$$

Or equivalently: $\text{Confidence}(X \rightarrow Y) / \text{Support}(Y)$

Lift > 1 (Positive):

Items co-occur more than chance. True positive association!

Lift = 1 (Independent):

No relationship. Occurrence together is pure coincidence.

Lift < 1 (Negative):

Substitute items. Buying X makes buying Y less likely.

WHY LIFT MATTERS

Tell students: "Even if Confidence is high, if Milk is bought by 90% of all shoppers regardless, Bread doesn't really boost Milk! Lift tests for true correlation."

Support vs. Confidence vs. Lift

Metric	Plain English Question	Formula	Range & Goal
Support	How common is the itemset overall?	$P(X \cap Y)$	$[0, 1]$ • Higher = Frequent
Confidence	If X occurs, how often does Y occur?	$P(Y X)$	$[0, 1]$ • Higher = Reliable
Lift	How much stronger than random chance?	$P(X \cap Y) / [P(X) \cdot P(Y)]$	$[0, \infty)$ • Value > 1 = Positive

UNIVERSITY EXAM HIGHLIGHT

This table directly answers the 5-mark question: "Define and compare Support, Confidence, and Lift in Association Rules."

Unit II Grand Summary

Pillar 1

Bayesian Networks

DAG representation of uncertain variables. Exploits conditional independence to avoid huge joint distributions.

Pillar 2

Influence Diagrams

Adds Decision ($\square$) and Utility ($\diamond$) nodes to Chance ($\circ$) nodes to compute actions maximizing expected utility.

Pillar 3

Association Rules

Unsupervised discovery of item co-occurrences ($X \rightarrow Y$), evaluated quantitatively by Support, Confidence, and Lift.

UNIT II COMPLETE

Congratulate the students on completing Unit II (Bayesian Decision Theory). Direct them to the final slide for semester examination question patterns.

Important Questions Bank (BCA Sem V)

2 MARK QUESTIONS

1. Define Bayesian Network.
2. What is a node in a Bayesian Network?
3. Define Influence Diagram.
4. Name 3 nodes in Influence Diagrams.
5. Define Association Rule.
6. Define Support and Confidence.
7. What does $Lift > 1$ signify?

5 MARK QUESTIONS

1. Explain Bayesian Networks with an example.
2. Explain the components of a Bayesian Network.
3. Discuss 3 node types in Influence Diagrams.
4. Differentiate BN vs. Influence Diagram.
5. Explain Support, Confidence, and Lift.

10 MARK QUESTIONS

1. Explain Bayesian Networks with diagram & conditional independence.
2. Explain Influence Diagrams with Chance, Decision, and Utility nodes.
3. Walk through Market Basket Analysis with sample calculations of Support, Confidence, Lift.